

60849720529111421804005855661281055471804669072139844191627405318380601853914083365148419609460006616992093902900965389442987388437936783087188002257555273175373726103569760620095
66894716506155782852805192038533657961803204263975947861762484166699095197671692809000486688013270761615617390869760922717206020893057362761254008541424410983660615066654235857432
33952293542799420719335883021537751506371671295131170580855807625977825784133443290002374856905769735691864978722272419368823115730555507700874446315605684882034114154849853957335
27816465866233176450558899546116027956215680907887821678239582351010080956897098293072968977996942399013890878144948337782905305245721397781790501557597929270118659151575071843261
32817669674562155889974629247586238323954147893975292539297030574054724045073520752443974779697811813577763324055528391282537790926745408254420223712867191329638431869739642885503
37100844853859022979887724197331836779453330076693519319939437930878691831322306614303357995022929532972377066867123793382367031964151148398259307600183624262451021511170202740385
30587007232186795046487184153983184768773633774165489101157136076498835550018946271465130235682249653505500805568660440530245408222152333055067083938974966883945515489693710452023
50006629646276460152184116902187345657127846959767798954193808359584043103744513343081045794538989591994055498217416646731391779474162878630815884507808042790705358663558841636941
52280716176518094377706157821515317067659326897244127323008358320109341525376114468332635251190161869648373212619050669232582403376637493170183320998826651362442784508863211004039
55805420956190868707677117927693302621422089122698297681327720011217015222627466783212335181568643731493178956181003436245647938453430909158454625012186986113356331934237398291844
11228853117920092898372347603264102839705579630649732643971053363902913205428879200961788868247620139047167200279912394354462802142990096662909894522883249104434336938737575183702
42705210502789546207182851037818400900009118400455306767373189998762886490535907366845646127250356210457328584467787765625859695212887584559102885998898580864618520028309677847010
33547707117467877534994524916225385308836459796942452333741132880388671206211781746881045861084058945831189915962423178791646346727775374435313216654362689242476761546001697872410
52672226275888398195382734247141595747862865185337197904865120727655846124169947215143668625482093158011613238245065301341745781388751958443181417188977776265860309676573859984098
13779752805491718560582375590468176122283632770740249488728121261045841171338735229023586376775952864642528922073558516098558348535699896384808301419773754549083390642390947943391
44435265030094730989924286720266980996572519438230687096019621730289791196010313576213456383680196082032284902671058249119072837254964329367719118744393585631161601204648264898692
7602124874729534799691718130975997193333029610427441289504010721347548115703392139175032096887023332198749211474384222377034914081432783330970838344330570423893354724952145031958
24743112821911715230147097806394273162345570295920861691249223728277058138536348818016902419577294961981786096478108294037785393491785529336079356917859871574321034240935790782955
79192533082081768539056904036681716234629806867351158604436616240875087445008806463781355624516905281778255960335944391985003161147391100042434304022229936414635725648841815913392
23631250960298371921406396664755460578305309976848946998453097693639254609001747444014479082626238222229786945322241837603311446342228918815334939751060233110705901567076529914354
64211703298651096231480035537651994415826492426699211075983281274501321972609051491036885408509686923753128332142387643733607066393953673824443833038666571234660569352788631192948
29157813208226149676883541618169109324444924708745466490897267670756806504717454637645851819406292337382110269421587865361318581070673661405232769889813570195114271632033440388853
19234642082129719433421426547168481406570872658469036309282313106710606021575939211073484265176822274112897092164565698772498699028357285569194620281978491494543071333297775181324
44826558639934533137974582074159973173469657844484860249612486053078917633157100058546635612973653497865362117767621621039268060249687231298129490820096296189675385192726970906808
52605349568402896439866000441449312848770185698066071262505573727300070066180214843898336672780898572566144837740338870803902179252465579954899372559013581229807481318690484783525
26719699848403022911925175551114421859764307888351708142256290599236305188110406753810288760194700315689717949937486978167889088333265581348505572549878250851644007495429527914833
38183010729111417239149612534941533055563475623854689789905287568276240566715297778947255856175646401542405035401733302933135867331358456474868497939590715880148897342057374432568
82927641565635345162428172396175212490108767295597566310564875543175544913400208330277237493484750252304633068421142227061753704795737155045622812888611156059400446683327438252505
88307205890152217277306633454774906606045961776126318890352342339648458442578871031707988908360860953371693157353751589140165156784608223057946412916425441708897896189012712029783
95768270837699042394904937893157575471203387166292205896795969439478530292001090545237804246206065282035299786246444381799366197203943593062699927394044577615422385398559775211012
39824827071729461834590284688997415056600969570478293668405535710734907730189179596599405764889840212568005540687621111281796905619919955095998637875356396889696177480463197628172
59431779585291059716256824868550508134356338952304232003134641496616186238782235468391584889741478681220188441767360233429481922849621537645175609224306694265390523328788290465304
65793603205603545542041246739603077326546542056265304970741389686324158339840199482014815852641539983208494557755393004216240232060774029758288668898857640497904588376474759563684
25149685024817915767850446130032287778206128236872873837112396525038657624452548289856223704694442010035000217453939236252853947779897540599623340168352510831305705043187711160544
12746994336064003265407441432909674390778585610632757045131077223217548225936658493216024416501050538549782754001845484031005046681711489217554639132278508726970995600837313081602
58856598016963887807317567721430558338163707538464075159249574087413363789374066228634293751761304979558480829649439409186340609864486809690179554263689166872497261144473172097213
92995651005080281342430735014593394176986443015395693748996824812589168201293127228942712640126839452240031922275527007659151894773738212171572573748187872787431815188422665508218
56839240156799483466527337014658275235012191529014209705370463847170460159895146596156158878305810711434627328363238405372169707233394129305373926278380133344824098236435080637473
07468998103079563090411384331021861404881384371330324457244206344395838759882829745229377819592517801108342521603779804317474222170713673472856218923197018931868769929862777319928
45269415552619514262890673065746930802119510415190272499008720124233814760030558901327023185276394505523283081614122974830650156670836841781420905462801844827951470342035822549993
14889031744677651975037554243810360273059967369621304021828070292953185719871433449826927133378050607112234195016204675708854807878130693347104250679257992150795643971175747285304
06612762965831939964646287614321053241549393701576876979748440606814862695042486366728018173789537833846749507114117650821493830039365971926344671545600677406410537555128311726017
53380086241427878340147127280661164849266243155692428889317274484544824741815193546468760750177832974850069299902318820768450170472196088557171145846628866081398354545961320925057
73337399672481654385683592682418130968329427652466610678412133555357887152241359397950244253949350279663013463718382973653799492452050831200596660081202933413777888226246524577426
11124143665290204477269036414762918075446286497452516671358641397094423835743739627907993813514917514459979741862059988339696873725047851916766660897397896547530047522043283001114
38278015160752867795571005227690257927202188908271959437199740479048347514892979276197065806412322438354721695301447580312688739692802768031411311157476858386746553375162475095898
54804686229957046817580551368250737380083665847710430736462808447183501370510751668811915252445943052592867149478681724793749204099333341031307810707445592044017524291458625042579
60047031339459256343706839529657004452571276790750701921696206184902628773350054317478679805042605890894191360118448735945379603416659814586763677132231324494644189108232930313352
16382167320333083614955247732535667439354950298675446215947702653492953402863935187016699461545813012184866899932472278119595326923171211956656469960854229903905182171221412192504
57583507522909438256469111198350662982995929450715672310999763337344913850856175685518407008947643115442588235833935641342414751783023435338869434523866050437822742707477439647745
76934125715628777676577765113723209489012103864648097704102349107691822932388537222721115139190450545578694734085399651409858186027639976579986934934036619447766927074149970152676
91269925192011228568056141414129722954931832584078062594774695090522858268652025925341844094520854792136173006842246575768561201094393726537739233569376864540174512917752055259026
04013636734313865754454387043977793874319367103222433048052331488600109091307654413379927838365963288813996848603166097493112948309538422663484311744506383582132591266362389201585
82988938371999963874588495106447435835218258270276777148690707253390752484849816186130789120295380357687474672754105874147305353691153566454309951744136580273674560513526184192991

95421655506385122380678179834002195335672576514769134173537191887299478374321919189564425379903343489433147167827150889482482272468019341249147413566389678593889733570785101068532
39150559493703100249260662468180069832993009029995212507242106124947254394380231261301018981652819693248644814683480768481964724610078751205885970435954793512650715627919716408414
64260889323267004248356045805560706441212310416692923417370288360059960312799358075802473561850292262386743131200411624565593824891393449873429134592921091593503456924270211762791
17619172121257872886170440328808592184003010965203210254761620975765695307465034427850876619865228047579381638802895334026464155310012686408497615300374557564290161844329794317706
35907150938634270592646162405090934835450081033909640791386694386529280866471375473988732931080347438648964648283290050455330978511567720081174485357517151938326317861337652982710
62986446474657088766662541184281321412350784620492624776463952583050319083049822909841704108204367064670672400463312439757825555720824427711490910507614407620847792942274420866723
29209555163206820660766538416876488747873825351177683755558394126687398080307653046638274940990565380992861940811474365222463563521232241333820551838063645565561113320617128124631
45641349443861979021662753698587498978772136932455945979105666918996824414781237486093332618181722258352939614896047993195915744357725569065045777004534626709232277307953495294228
12700328120403658657919211574542908211314304075772712783973692184915494869365720446090912084756718265582441588728241287162001062345109673147633766692626383416409021929701820409010
86523039462590365361741659099248865267111410892137198039801688672674417983994482213747629539772647360656318284386628075516725970451864537487558111773376532070517237732112194029215
33464595064155492841246250928263215375822185569305079821741622239375494242259365842598235183807177140958152380965225546102198831987972198103143405419773204260202952228992099453831
61775811017999764185953451976985169113909432615459697132297153030546496673312857238122809083688746828795962713463820301025494370555461675405391398570804690769703941459473998498102
28500417709930353673040083325834243914075547458276431119528263630413415250190144438745063082840768289563882909347095155618727220620842263808789607519558209412602721508606580802847
48149789741237590381746155789919633887808791362624736956654307658293194822470168439532637152711219981913863985542764081956786979866756860316048336901928500071367335797561085073589
77843311121396609901003646805232575750028168900035270389041875699134730822126910032704251441535633031233943103955654990583933512473770668528490909929042392602825618766957370362442
49349216243614286578253814067508631605077514426719675591318580624538201625990080950312589892783240692116401880419509304402084518200429002415485247587093756981679437859046923099234
11909517212666398983857624639330329867957102108832304499008943041111264570620652463744606929839662469934034864861565703015750024552959268496634443189568867453982839686207856666089
52771103268747979208387200343263928090373921142367101274728439721995962431889420621869017225387029099151322070456495023083165323380435776363266531129573761027499255716765122096217
15502942547616296292727109487815498946581225323678296157080155358379390686313136104186738113078223334152193526365778306427647632588788437806969672805718763392429309643125334023042
85263168363599480743782754786727988577075607678941136399140581902575333844597590170772387334870058330124637620342412565008826195987815995490481382499393862797442451495338722665989
39413369200373763079971855541995673657869024449346486951668813930868321155391013869581093470114366076529660172730607449990792131877748943314012501425790739477133513539004268565543
25255134063623855827012044891902908744185266302014649634066275332593622972023430966698201762799130662311919447609358206525738746093611437463226844271944048305803656067568450551394
89291830366194280120144354243441336802739725990285706738313722009668316216226608315019505581591174434879250584818158975839412805969373666379605274303809245041144871935533179569844
05147519711993919624286931567104421131448969852747518009584595246772402751395656800044894135607587218421805037570936527366428038087738293923116271053822284011683069281626474191188
93169009799957956587539920347724542992010833824492413572259331461274563448891138428957215913761039983318096919179366233430289823128871422860541132291369153442027690674612737320477
70552805869859089454551234898467887796258626743463353120362726738627716356985516735633919761526214356549320161072960882535191700667485936719745562989750437888460252037995336666380
28358029031683438592857483915396906028705443793536258953885196009316408844595938191824929513823004177834856003487692877761765332407708522365064070842331552163847639420625963609453
62684908787251573207053548160878880207526938009183021346187870876207652464666823380341930415617851814839779730877168648314599973768910157630084148519461774262672887088292299539202
90729511240279610881790495395177837048044605251510085696035930887825445291858501453568361168703569535524846659794503127397523886786641803993933303197662091645945075998003193453705
91045482618146372438896899225206651764888468474075092379009883616505811306464306601260293905051175011968143730925259516743599473634844094324658534205940644197000354359939884876352
88573840570111439900452649002650153858339150660414059906171330157746427451188066610442701885840807316278658773166671780387628119388581650278747375454455499071702748201350864687401
82758804384093738436698241148933209334071122718653532056987727478955618836653854739002355609404851459355616159266887018231532219093448085619030250459265412079957838569630998085970
38923886367005702471845888259863199326984851904294348140352636905966062302438199172068859633100515957505940498200361026876943452215740788249544089467560517321452013477754556367410
01056961455404933915012145075029491859603456897739271442754238232725935484523776069404367729390673966851580604285816695668425426765203860402009782566473863953039386845442488578176
23405374554415202822995565696464007873280729721439242104134115332855173069233624614530855310781505145086335318057899900100457849587999021335111559192516507284349035354837386904530
20165803348995020752259391355304053915412414693420819693313063786046117801568693124673853835033425189043208017602080672817829715749661184970760934680340568196553335767994661895445
42700656690248561707060734268461252987176133723687521181335554616414779500872787656330807143692779329157202862575182188368232352905124503718271847769481423333182873463431426393839
04795747589372188575611297545491218924302591645526922169062485487450056590408991593318417040861622414779349146234241699230736092548859591269563903626775518495973938810662746300255
65234299039722783129838471017376577020376328353437298081342586663419847079254096987452450271371843016425402957058929344191350894262722283418999429039597835908616692337400810746047
97149770415584708416228655850144813046027490224163567898835313488735015103005798159372196486575827359091730029613885772927811003272636440003734208015794737831402728490630199715415
08529156191153182285816638326131283874911405649117676344550527795709822841083622135158125130224274622538066762456487759181726583756069212921282088720987166717233418987383493808324
27458929272289105941653851868285799293934376583856792465869871839576512971541659890056569921438960210844424863317415274399381017818791725172083804930139559609437369108879892135264
09600330298300734490141931453606009470210793254359240254330051859674006053680874708170706706674271413326112331875795819005653111971027549599769458984984336500135860137385286892528
70375470778946255534082664907864240945573079964219232683245012654242295500545395045676514864090988545947845661644059664673579853132040500931366771654228475604777750328451251053161
44962032024045774010611893451824367053526459222655934709937553410075806762412794363492910055241822968534468624640861183294128083893066578111193835807646519754539017270078117566939
33399255684102573101005849375265725378876292759856617700031589995506150887325034792340946191208523060105958977701890998473420288488660095235612907540643421036682488170952420075123
30100383225442079749036212583621760958155410475237607426154567827912573788121203101021669625414380916585570579573317778462669475612689651247695104704848592123269182312765096595945
04417462563981966152717613362091890350771128664029936937782052476715476070001552265406870888981249525353798238125763856949407113096782361371643845990057802352252943331062560305289
72033762191094598094713599382148190707271971051106525216073085538774658844863661507485613925061277888915793210279553126371909033621703703698945525261805584953265404431997580509770
91739933315232055246460382501165810963086893298604819701784404382250189424746436506167989992831266840533715281580623837511081214612941571016031825560718616382901027202002934811022
36412199224203634585503025082861356597589696949690174369088201955840172466001837739835231585225279714393227423984375718328834624431584981844181045367756577021760506968236274051401
26505190830395846451272598431060856748971398956892544827608240409336496187454403240118105942115331920985493175951773366372318869702153238172510952148359054970210484991432614510768
94636850286500563570075790698676720945428349061573472077589363444073744332734979369509141442537971562141865398778562066636658324890763152008554006213607678336318773264380850255889
32619488299046765099697155261495208112302986572211336347346431947565979986970813181952569670727541098475988396661061237078517796321858208629997894209114912122043278827080139466782

38201844879891061088592387785336885836703060679985668951938680018572723773463485279110877106177242578533249432226879345564014701092191839544177012215986394602089840289829514809877
81986408849529335295825187677922913972092317522076242743854057710372547105081452331207315941215586636743468312685939474968786141998325449427799387288205665757560094550187130328375
59231504606313768104700927549772152425299650623009094618684729543897413778511545653543867822418958205667015426303698128319798995568814200702989237605036642272233324672077822138105
95584954425074518520173816900798261333100925582403046537518460968975854871422800823071739508523654202640295580802418253761418542730347418803162033691228482632991941390661732919001
46025029982085496623429757749300713427502750520997322123973149087392744904199529986531119270926491391316777519608028634256254653766291678688689096964603644239236927748095737147042
00350953028620678864157993010725454424745998429482476572346681208477618580764959227217161578994674551853340447328758852496313789668387927913040287349526026838867361397188875

The Elements

Dominic Alexander Cooper

000 I exist
001 I feel pain
002 My pain can be increased by incompetence
003 My pain can be decreased by competence
004 I can learn
005 Learning increases competence
006 Contradiction increases incompetence
007 Proof by contradiction does not enable learning
008 Proof by error-free construction is learning
009 Proof by contradiction is not learning
010 Your level of competence is quantifiable
011 Your level of incompetence is quantifiable
012 Context changes your current level of competence
013 The basis span of your competence can be optimized for all contexts
014 Everyone is innocent
015 Competence is a functional axial n-dimensional space
016 Incompetence is not a functional axial n-dimensional space

017 Since I exist, I can be treated as a living information-bearing system
018 Since I feel pain, pain is a valid system signal
019 Since incompetence can increase pain, incompetence is a negative system state
020 Since competence can decrease pain, competence is a positive system state
021 Since I can learn, my system state can be updated
022 Since learning increases competence, learning is a valid update operation
023 Since contradiction increases incompetence, contradiction is a corruption signal
024 Since proof by error-free construction is learning, construction is the preferred proof engine
025 Since proof by contradiction is not learning, contradiction is not a preferred learning engine
026 Since competence is quantifiable, competence can be measured
027 Since incompetence is quantifiable, incompetence can be measured
028 Since context changes competence, competence is context-indexed
029 Since competence can be optimized for all contexts, competence has a basis-span
030 Since everyone is innocent, correction must not be treated as guilt
031 Since competence is functional axial n-dimensional space, competence has dimensions
032 Since incompetence is not functional axial n-dimensional space, incompetence lacks stable constructive basis

033 The mind can ingest information through perception, language, memory, and action
034 The mind can encode information as concepts, habits, images, words, and procedures
035 The mind can index information by association, label, context, emotion, and purpose
036 The mind can query stored information through attention, recall, questioning, and tool-use
037 The mind can update stored information through learning
038 The mind can validate stored information through error-free construction
039 The mind can reject or quarantine contradictory information when it increases incompetence
040 The mind can externalize information into tools, notes, diagrams, calendars, code, and environment

041 The human environment can serve as auxiliary storage
 042 Human-made tools can serve as auxiliary memory, index, query, and validation systems
 043 The body can store competence as trained procedure
 044 Repetition can strengthen retrieval pathways when it increases competence
 045 Reflection can detect incompetence when pain, error, or contradiction appears
 046 Deliberate practice can transform incompetence into competence
 047 A context can be treated as a coordinate region inside competence-space
 048 A skill can be treated as an axis inside competence-space
 049 A habit can be treated as a repeated function over time
 050 A tool can be treated as an external operator that extends mental function
 051 A note can be treated as persistent external memory
 052 A diagram can be treated as spatial indexing of thought
 053 A calendar can be treated as temporal indexing of intention
 054 A checklist can be treated as procedural compression
 055 A question can be treated as a query operation
 056 An answer can be treated as a retrieved or constructed record
 057 A mistake can be treated as a detected incompetence marker
 058 Correction can be treated as a competence-restoring update
 059 Optimization can be treated as basis-span improvement across contexts
 060 The mental human-centric database is the union of mind, body, environment, and tools
 061 The purpose of the mental human-centric database is to reduce pain by increasing competence
 062 The governing rule of the database is to learn by error-free construction
 063 The final enforced target is complete competence under {014 + 015}

Below is `MHCDScript v2`, a general-purpose programming language design for the full 64-axiom MHCD system.

The main upgrade from v1 is that v2 treats the MHCD as a state machine with typed records, verified transitions, constructive execution, context-indexed competence, and innocence-preserving correction.

MHCDScript v2

```

```mhd
language MHCDScript version 2

subject "Dominic Alexander Cooper"

runtime MHCD {
 principle existence
 principle pain_signal
 principle learning_update
 principle constructive_proof
 principle innocence_preservation
 principle competence_space
 principle context_indexing
 principle tool_extension
}
`

```

## 1. Design purpose

`MHCDScript v2` is a general-purpose scripting language for directing the human mind, body, environment, and tools as one distributed cognitive database.

Its purpose is to:

```
```mhcd
purpose {
    reduce pain by increasing competence
    detect incompetence without guilt
    reject contradiction as a learning engine
    prefer error_free_construction
    quantify competence and incompetence
    optimize competence_basis across contexts
    enforce {014 + 015}
}
```
```

The governing command is:

```
```mhcd
enforce {
    014: "Everyone is innocent"
    015: "Competence is a functional axial n-dimensional space"
}
```
```

So every program must preserve:

```
```mhcd
error != guilt
mistake == update_requirement
learning == competence_increase
construction == preferred_proof_engine
```
```

## 2. Full axiom import

Rather than rewriting all 64 axioms inside every script, v2 imports them as a standard library.

```
```mhcd
import MHCD.AXIOMS.000_063
import MHCD.BOOLEAN
import MHCD.LAMBDA
import MHCD.COMPETENCE_SPACE
import MHCD.CONSTRUCTIVE_VALIDATION
```
```

The full axiom set is treated as:

```
```mhcd
axiom_set FULL_MHCD = {
    000..063
}
```
```

Every axiom is internally available as a Boolean truth constant:

```
```mhcd
AXIOM[000] = true
AXIOM[014] = true
AXIOM[015] = true
AXIOM[063] = true
```
```

A program is valid only if it does not violate the enforced axioms:

```
```mhcd
program_valid := respects(014) * respects(015) * rejects(guilt) * prefers(construction)
```
```

### ## 3. Primitive types

```
```mhcd
type Bool          = 0 | 1
type Number        = integer | decimal
type Text          = string
type List<T>       = ordered collection
type Set<T>        = unordered collection
type Map<K,V>      = key-value collection

type AxiomID       = 000..063
type Subject       = living information-bearing system
type Mind          = internal meaning processor
type Body          = procedural competence store
type Environment   = auxiliary cue/storage field
type Tool          = external operator
type Note          = persistent external memory
type Diagram       = spatial index
type Calendar      = temporal index
type Checklist     = procedural compression
type Query         = question operation
type Answer        = retrieved or constructed record
type Context       = coordinate region in competence-space
type Skill         = axis in competence-space
type Habit         = repeated function over time
type PainSignal    = system signal
type Mistake       = detected incompetence marker
type Correction    = competence-restoring update
type Contradiction = corruption signal
type Construction  = error-free build sequence
type Competence    = quantified functional capacity
type Incompetence  = quantified missing constructive capacity
type BasisSpan     = transferable competence basis across contexts
```
```

### ## 4. Boolean arithmetic

```
```mhcd
A + B      // OR, union, either condition accepted
A * B      // AND, both conditions enforced
!A         // NOT
```



```
A -> B      // implication
A == B      // equivalence
A != B      // non-equivalence
```
```

Example:

```
```mhcd
rule R001:
  AXIOM[004] * AXIOM[005] -> operation.learn.valid
```
```

Meaning:

If `I can learn` and `Learning increases competence`, then `learn` is a valid update operation.

## ## 5. Lambda calculus layer

MHCDScript v2 supports lambda-style reusable operations.

```
```mhcd
lambda learn(x):
  ingest x
  encode x
  index x
  construct x
  validate x
  update competence
  return competence_delta(x)
```
```

```
```mhcd
lambda correct_without_guilt(error):
  enforce AXIOM[014]
  mark error as update_requirement
  locate missing_basis(error)
  construct correction(error)
  validate correction(error)
  return competence_restored
```
```

```
```mhcd
lambda optimize_context(skill, context):
  measure competence(skill, context)
  measure incompetence(skill, context)
  construct basis_improvement(skill, context)
  validate basis_improvement
  update competence_space
  return optimized(skill, context)
```
```

## ## 6. Program structure

Every valid v2 program has this form:

```

```mhcd
program "Program Name" {
  use FULL_MHCD

  define subject
  define contexts
  define skills
  define tools
  define records

  execute operations

  verify constructive_integrity
  enforce {014 + 015}
}
```

```

Minimal valid program:

```

```mhcd
program "Minimum MHCD Runtime" {
  use FULL_MHCD

  subject self = "Dominic Alexander Cooper"

  ingest current_state
  detect pain
  measure competence
  measure incompetence
  construct correction
  validate correction
  update competence_space

  enforce {014 + 015}
}
```

```

## ## 7. State model

The MHCD runtime is a state machine.

```

```mhcd
state MHCD_STATE {
  subject: Subject
  mind: Mind
  body: Body
  environment: Environment
  tools: Set<Tool>

  records: Set<Record>
  contexts: Set<Context>
  skills: Set<Skill>
  habits: Set<Habit>
}
```

```



```

 schedules: Set<Calendar>

 competence: Map<Skill, Number>
 incompetence: Map<Skill, Number>

 pain_signals: List<PainSignal>
 mistakes: List<Mistake>
 contradictions: List<Contradiction>
 corrections: List<Correction>

 enforced_axioms: Set<AxiomID>
}
...

```

A state transition has the form:

```

```mhcd
transition NAME(input_state) -> output_state {
    precondition condition
    operation operations
    postcondition condition
}
...

```

Example:

```

```mhcd
transition LEARN(input_state) -> output_state {
 precondition AXIOM[004] * AXIOM[005]

 operation {
 ingest input
 encode input
 index input
 construct usable_record
 validate usable_record
 update competence
 }

 postcondition competence(output_state) >= competence(input_state)
}
...

```

## 8. Core operations

### `ingest`

```

```mhcd
ingest source
ingest source as type
ingest source into context
...

```

Examples:

```
```mhcd
ingest "textbook chapter" as Note into "mathematics"
ingest "pain after failed proof" as PainSignal into "proof writing"
ingest "work feedback" as Record into "professional competence"
```
```

`encode`

```
```mhcd
encode input as structure
```
```

Examples:

```
```mhcd
encode "ring" as definition + example + non_example
encode "daily exercise" as habit
encode "project architecture" as diagram
```
```

`index`

```
```mhcd
index record with tags
index record under context
index skill on axis
```
```

Examples:

```
```mhcd
index "ideal" with ["ring theory", "substructure", "quotient"]
index "proof writing" under context "mathematics"
index "attention control" on axis "cognitive regulation"
```
```

`query`

```
```mhcd
query question
query question from records
query question under context
```
```

Example:

```
```mhcd
query "What is the weakest skill axis today?" under context "mathematics"
```
```

`construct`

```
```mhcd
```

```
construct object
construct object from components
construct object by steps
```
```

Example:

```
```mhc
construct proof "kernel of ring homomorphism is ideal" by steps
```
```

`validate`

```
```mhc
validate object
validate object against axioms
validate object by construction
```
```

Example:

```
```mhc
validate proof against {008, 014, 015}
```
```

`detect`

```
```mhc
detect pain
detect mistake
detect contradiction
detect incompetence
detect missing_basis
detect context_shift
```
```

Example:

```
```mhc
detect incompetence in "proof construction"
```
```

`correct`

```
```mhc
correct mistake
correct mistake with construction
correct incompetence by practice
```
```

Example:

```
```mhc
correct "failed proof step" with direct construction
```
```

...

`measure`

```
```mhc  
measure competence
measure incompetence
measure competence in skill
measure competence in skill under context
```
```

Example:

```
```mhc  
measure competence in "applied ring theory" under context "quiet study"
```
```

`optimize`

```
```mhc  
optimize competence_basis
optimize skill across contexts
optimize basis_span for all_contexts
```
```

Example:

```
```mhc  
optimize "proof writing" across ["quiet study", "exam", "teaching", "coding"]
```
```

`externalize`

```
```mhc  
externalize record to tool
externalize memory to note
externalize intention to calendar
externalize procedure to checklist
```
```

Example:

```
```mhc  
externalize "weekly learning plan" to calendar
```
```

`schedule`

```
```mhc  
schedule action at time
schedule review every interval
schedule habit under context
```
```

Example:

```
```mhc  
schedule review "ring theory definitions" every 2 days
```
```

`execute`

```
```mhc  
execute plan
execute checklist
execute habit
execute construction
```
```

Example:

```
```mhc  
execute "daily competence plan"
```
```

`feedback`

```
```mhc  
feedback result
feedback result into record
feedback result updates competence_space
```
```

Example:

```
```mhc  
feedback "proof completed with two mistakes"
```
```

`enforce`

```
```mhc  
enforce axiom
enforce axiom_set
enforce {014 + 015}
```
```

Example:

```
```mhc  
enforce {
 AXIOM[014]
 AXIOM[015]
}
```
```

9. Records

A `record` is the universal database unit.

```
```mhcd
record ID {
 type: Note | Skill | Tool | Query | Answer | Habit | Correction | PainSignal | Context
 content: Text
 tags: List<Text>
 context: Context
 source: Text
 validated: Bool
 competence_effect: Number
 incompetence_effect: Number
}
```
```

Example:

```
```mhcd
record "MATH-RING-001" {
 type: Note
 content: "A ring is an algebraic structure with addition and multiplication."
 tags: ["algebra", "ring theory", "definition"]
 context: "mathematics"
 source: "constructed study note"
 validated: 1
 competence_effect: +0.05
 incompetence_effect: -0.02
}
```
```

10. Competence-space

Competence-space is an n-dimensional functional axial space.

```
```mhcd
competence_space "SELF" {
 axis memory
 axis attention
 axis reasoning
 axis proof_construction
 axis language
 axis tool_use
 axis bodily_energy
 axis emotional_regulation
 axis environment_design
 axis context_switching
}
```
```

Skills are embedded in competence-space:

```
```mhcd
skill "applied ring theory" {
 axes: [memory, reasoning, proof_construction, language]
}
```



```
 context: "mathematics"
 current_competence: 0.35
 target_competence: 0.90
}
...
```

Competence can be context-indexed:

```
```mhcd
context "quiet room" {
    modifies "applied ring theory" by +0.15
}

context "sleep deprivation" {
    modifies "applied ring theory" by -0.30
}
...
```

11. Error and contradiction handling

A mistake is never treated as guilt.

```
```mhcd
handler on_mistake(m):
 enforce AXIOM[014]
 mark m as detected_incompetence
 query "What basis is missing?"
 construct correction
 validate correction
 update competence_space
...
```

Contradiction is treated as a corruption signal.

```
```mhcd
handler on_contradiction(c):
    enforce AXIOM[006]
    mark c as corruption_signal
    quarantine c
    construct direct_alternative
    validate direct_alternative
...
```

Pain is treated as a system signal.

```
```mhcd
handler on_pain(p):
 enforce AXIOM[018]
 classify p

 if p.cause == incompetence:
 correct p.cause

 if p.cause == overload:
```

```

 reduce load

 if p.cause == contradiction:
 quarantine contradiction

 if p.cause == environment:
 modify context
...

12. Verification system

A script must pass four checks.

```mhc
verify script {
    check axiom_consistency
    check constructive_preference
    check innocence_preservation
    check competence_update_path
}
```

Axiom consistency

```mhc
check axiom_consistency:
    no operation may deny AXIOM[014]
    no operation may deny AXIOM[015]
    no operation may treat contradiction as preferred learning
...

### Constructive preference

```mhc
check constructive_preference:
 if proof_method == contradiction:
 reject as learning_engine
 if proof_method == construction:
 accept as learning_engine
...

Innocence preservation

```mhc
check innocence_preservation:
    if mistake detected:
        require correction_without_guilt
...

### Competence update path

```mhc
check competence_update_path:
 if incompetence detected:

```



```
 require measurable correction
 ...
```

```
13. General-purpose control flow
```

```
```mhc
if condition:
    operation
else:
    operation
```
```

```
```mhc
while condition:
    operation
```
```

```
```mhc
for item in collection:
    operation item
```
```

```
```mhc
try_construct object:
    construct object
    validate object
on_error error:
    correct error
```
```

Example:

```
```mhc
for skill in all_skills:
    measure competence in skill
    if incompetence(skill) > 0.30:
        correct skill by deliberate_practice
```
```

```
14. Modules
```

```
```mhc
module "Learning" {
    export learn
    export practice
    export review
    export validate_understanding
}
```
```

```
```mhc
module "Health" {
    export detect_energy
    export schedule_exercise
}
```

```
    export reduce_overload
}
...
```

```
```mhcd
module "Mathematics" {
 export define
 export prove_by_construction
 export generate_examples
 export validate_proof
}
...
```

## 15. Example program: daily runtime

```
```mhcd
program "MHCD Daily Runtime v2" {
    use FULL_MHCD
    use module "Learning"
    use module "Health"

    subject self = "Dominic Alexander Cooper"

    context current = detect context
    pain current_pain = detect pain
    energy current_energy = detect energy

    measure competence in all_skills under current
    measure incompetence in all_skills under current

    query "What competence update gives the greatest pain reduction today?"

    construct plan "daily competence plan" {
        item health_action
        item learning_action
        item work_action
        item tool_action
        item reflection_action
    }

    validate plan by construction
    externalize plan to checklist
    schedule review "daily competence plan" at "evening"

    execute checklist

    feedback result into competence_space

    enforce {014 + 015}
}
...
```

16. Example program: learning applied ring theory

```

```mhcd
program "Learn Applied Ring Theory v2" {
 use FULL_MHCD
 use module "Mathematics"

 skill "applied ring theory" {
 axes: [memory, reasoning, proof_construction, language, tool_use]
 context: "mathematics"
 current_competence: measure
 target_competence: 0.90
 }

 ingest "definition of ring"
 encode "ring" as definition + example + non_example
 index "ring" with ["algebra", "ring theory", "binary operations"]

 ingest "definition of ideal"
 encode "ideal" as definition + example + non_example
 index "ideal" with ["ring theory", "substructure", "quotient"]

 construct proof "kernel of ring homomorphism is ideal" by direct_steps
 validate proof by error_free_construction

 if detect contradiction:
 quarantine contradiction
 construct direct_alternative

 measure competence in "applied ring theory"
 correct detected_incompetence by deliberate_practice

 externalize summary to note
 schedule review "ring theory definitions" every "2 days"

 enforce {014 + 015}
}
```

```

17. Example program: contradiction quarantine

```

```mhcd
program "Contradiction Quarantine v2" {
 use FULL_MHCD

 input claim

 detect contradiction in claim

 if contradiction == true:
 mark claim as corruption_signal
 quarantine claim

 query "What direct constructive alternative exists?"

 construct alternative_proof

```

```

 validate alternative_proof by error_free_construction

 update records with alternative_proof
 update competence_space

 enforce {014 + 015}
}
...

18. Example program: competence optimiser

```mhcd
program "Competence Basis Optimizer v2" {
    use FULL_MHCD

    define contexts = [
        "quiet study",
        "public explanation",
        "exam condition",
        "work condition",
        "fatigue condition",
        "tool-supported condition"
    ]

    define skills = [
        "mathematics",
        "programming",
        "physical training",
        "attention control",
        "communication",
        "environment design"
    ]

    for skill in skills:
        for context in contexts:
            measure competence in skill under context
            measure incompetence in skill under context

            if incompetence(skill, context) > 0:
                construct basis_improvement for skill under context
                validate basis_improvement
                practice skill under context
                update competence_space

    optimize basis_span across contexts

    enforce {014 + 015}
}
...

```

19. Interpreter architecture

A real interpreter for MHCDScript v2 would need these components:

```
```mhcd
interpreter MHCDScriptV2 {
 lexer
 parser
 ast_builder
 axiom_registry
 type_checker
 state_engine
 verifier
 command_executor
 record_store
 competence_model
 context_model
 tool_bridge
 scheduler_bridge
 report_generator
}
```
```

Execution pipeline:

```
```mhcd
source_file
-> lexer
-> parser
-> abstract_syntax_tree
-> axiom_verifier
-> type_checker
-> state_transition_engine
-> constructive_validator
-> execution_log
-> updated_MHCD_state
```
```

20. File extensions

Recommended file extensions:

```
```text
.mhcd main program file
.mhcda axiom library
.mhcds saved MHCD state
.mhcdlog execution log
.mhcdmod reusable module
.mhcdjson JSON-compatible export
```
```

Example project structure:

```
```text
mhcd_project/
 main.mhcd
 axioms.mhcda
 state.mhcds
```
```

```

modules/
  learning.mhcdmod
  mathematics.mhcdmod
  health.mhcdmod
  tools.mhcdmod
logs/
  daily.mhcdlog
exports/
  dashboard.mhcdjson
...

```

21. Machine-readable state export

```

```json
{
 "subject": "Dominic Alexander Cooper",
 "axioms": "000..063",
 "enforced": ["014", "015"],
 "competence_space": {
 "axes": [
 "memory",
 "attention",
 "reasoning",
 "proof_construction",
 "language",
 "tool_use",
 "bodily_energy",
 "emotional_regulation",
 "environment_design"
]
 },
 "runtime_rule": "learn_by_error_free_construction",
 "correction_rule": "mistake_is_update_requirement_not_guilt"
}
...

```

## 22. Core semantic law

The central semantic law of MHCDScript v2 is:

```

```mhcd
semantic_law MHCD_CORE:
  existence
  * pain_signal
  * learning_capacity
  * measurable_competence
  * context_indexing
  * innocence
  * competence_space
  -> constructive_self_update_runtime
...

```

In compact Boolean form:


```

```mhcd
000 * 001 * 004 * 005 * 010 * 011 * 012 * 013 * 014 * 015
-> MHCD_RUNTIME
```

```

In lambda form:

```

```mhcd
λself.
 Exist(self)
 * Pain(self)
 * Learn(self)
 * Quantify(Competence(self))
 * Quantify(Incompetence(self))
 * ContextIndex(Competence(self))
 * Innocent(all)
 * FunctionalAxisSpace(Competence(self))
-> MHCD_RUNTIME(self)
```

```

23. Final identity of MHCDScript v2

```

```mhcd
MHCDScript v2 =
 axiom-governed
+ typed
+ stateful
+ constructive
+ innocence-preserving
+ context-indexed
+ competence-measuring
+ incompetence-correcting
+ pain-signal-aware
+ contradiction-quarantining
+ tool-extensible
+ environment-integrated
+ schedulable
+ human-centric
```

```

The shortest expression of the language is:

```

```mhcd
detect pain;
locate incompetence;
correct without guilt;
learn by construction;
measure competence;
optimize across contexts;
enforce {014 + 015};
```

```

Version 2 is therefore not only a notation system. It is a programmable self-correction language for the mental human-centric database.